

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 02

Introduction to Clinical Imaging & Imaging of Back

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 1: pg 7- 13
- Online Textbook: Chapter 1 <https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323934237000022>
- Pay special attention to the “**In the Clinic**” boxes for the clinical correlations for this topic
 - Blue boxes in online text
 - Green boxes in hard copy

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1032	Discuss the following types of medical imaging: Radiographs, CT scans, MRI and contrast studies
SOM.MKII.BPM1.1.MSK.1.ANAT.1112	Identify the major bony landmarks of 5 regions of vertebral column in imaging studies and use this information to interpret clinical signs and symptoms
SOM.MKII.BPM1.1.MSK.1.ANAT.1107	Identify the superficial and deep muscles of the back in imaging studies
SOM.MKII.BPM1.1.MSK.1.ANAT.1111	Describe the major anatomical features of spinal cord, nerves and meninges in imaging studies.

Lecture Overview

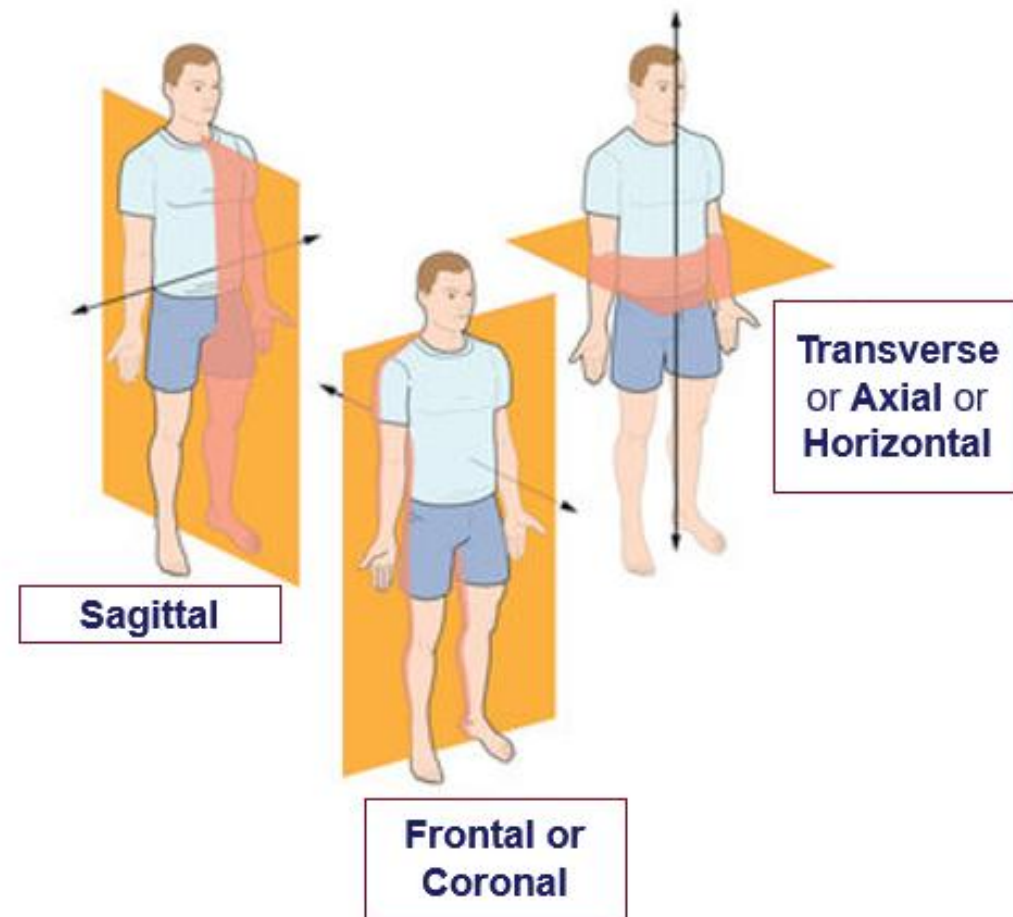
- **Indications for Imaging of the Back**
- **Body Planes and Sections**
- **Types of Clinical Imaging**
 - **X-ray Imaging of the Back**
 - **CT Scan of the Back**
 - **MRI of the Back**
 - MRI, T1 vs. T2 weighted images
 - **Contrast Studies**

Who needs medical imaging?

- **Clinical Scenario I:** A 28-year-old man is brought to the emergency room after falling from an 8-story scaffold. The patient is responsive but is unable to move the upper and lower extremities
 - **Clinical Scenario II:** A 55-year-old man comes to the Sciatica because of lower back pain. The pain started a two weeks ago after moving furniture. Initially the pain was localized in the lower back but is now moving down the back of the legs
 - **Clinical Scenario III:** A 70-year-old man is brought to physician because of back pain. He is losing sleep due to worsening of back pain during the night. The pain is affecting his ability to move around. He has a past medical history of prostate cancer
- **Indications for imaging of the back**
 - Trauma
 - Progressive Neuropathic Pain saddle anesthesia, bladder and sphincter dysfunction, progressive motor and sensory deficit in lower limbs, sciatica, and gait disturbance.
 - Congenital Defects
 - Suspicion of infection (Spine or spinal cord)
 - History of cancer
 - Fever
 - Progressive and prolonged back pain

Body Planes and Sections

- Always orient yourself! Start with the basics → Anatomical Orientation
 - *Face looking forward; hands by side palms forward; feet together toes forward*
- **Body Planes**
- **Frontal (Coronal)** – vertical plain dividing the body in to anterior and posterior parts
- **Sagittal** – vertical plane perpendicular to coronal plane, dividing the body into right and left parts
- **Transverse (Axial or Horizontal)** – divides the body into superior and inferior parts
- **Commonly Used Descriptors**
 - Anterior (Ventral) vs Posterior (Dorsal) – relative to the front or back of the body
 - Lateral vs Medial – relative to midsagittal plane
 - Proximal vs. Distal – closeness structure to its origin
 - Cranial vs. Caudal – relative to the head or tail region of the body



Types of Clinical Imaging

**X- Ray or
Radiograph**

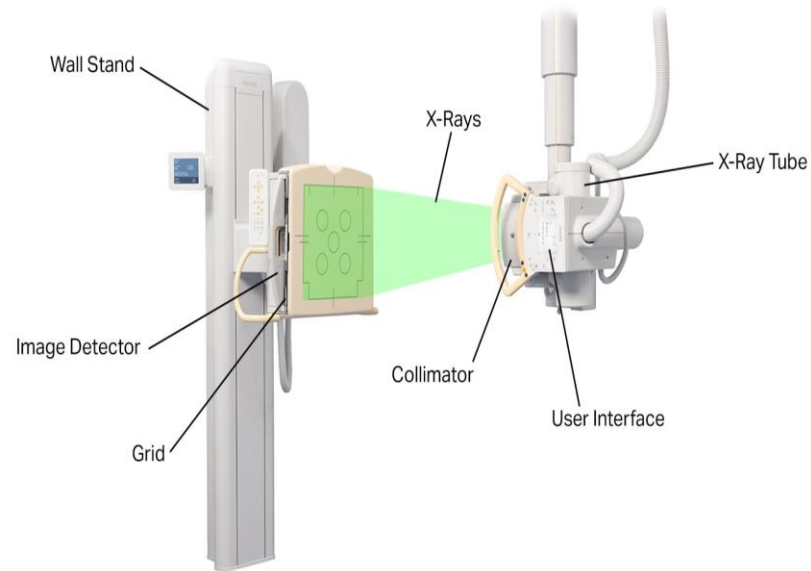
CT Scan

**Contrast
Imaging**

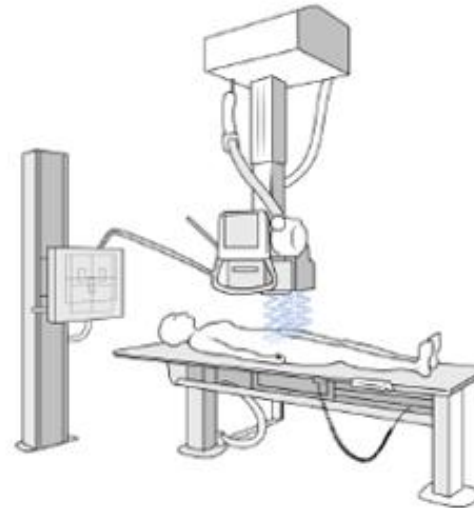
MRI

X Ray

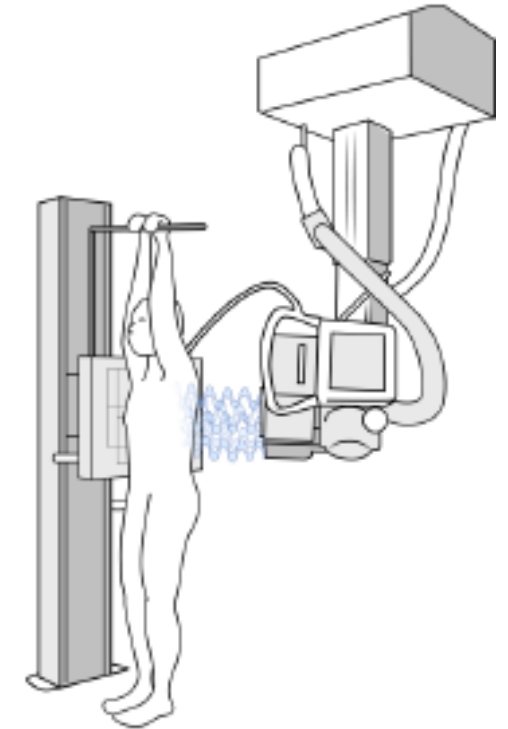
- X-Ray beams are generated from an X-ray tube complex
- Rays pass-through lead line shutters directed to an area of interest
- Positioning/Orientation of patient is important:



X-Ray Machine



Anterior to Posterior

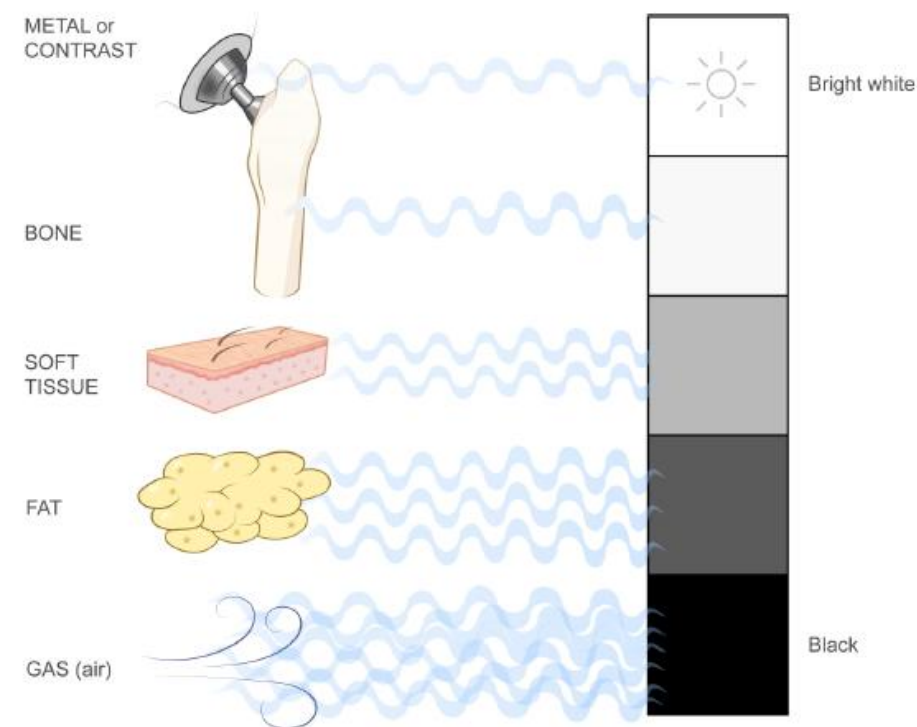
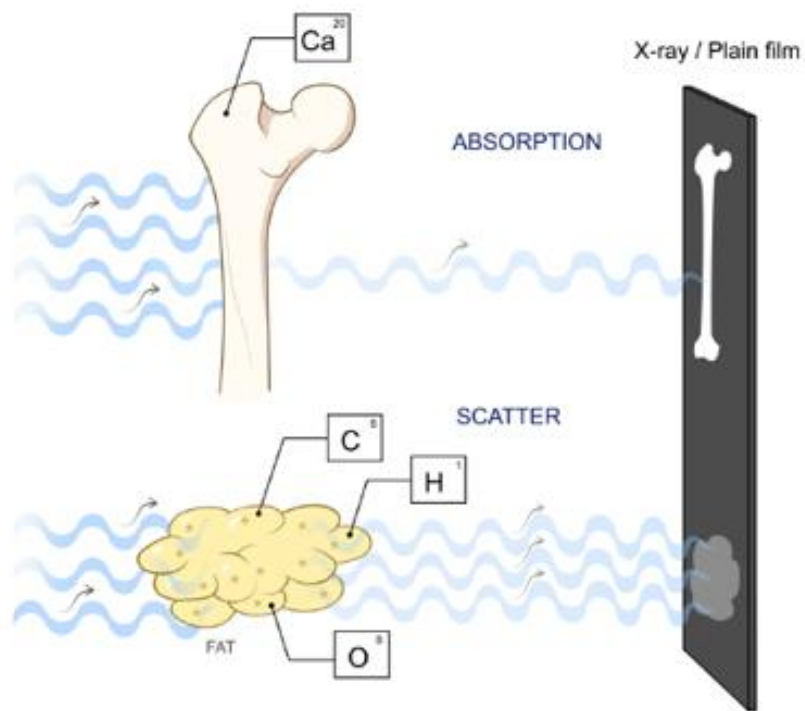


Lateral

X Ray

- Image(Radiograph) is generated on a photographic film or cassette is based on exposure to X-Ray beams :
 - Features of the radiograph depends on the tissue **attenuation** of the X-Ray beams
 - Attenuation is related to the density of tissue or implanted material
 - Attenuation increases of a structure /material increases density**
 - Metal > Bone > Soft Tissue > Fat > Air**
- Images on the shades of grey color spectrum
 - Less photons pass through structures → appear **white** on X-Ray
 - More photons pass through structures → appear **black** on X-Ray

Best for viewing
bone especially
when rapid
results are
needed

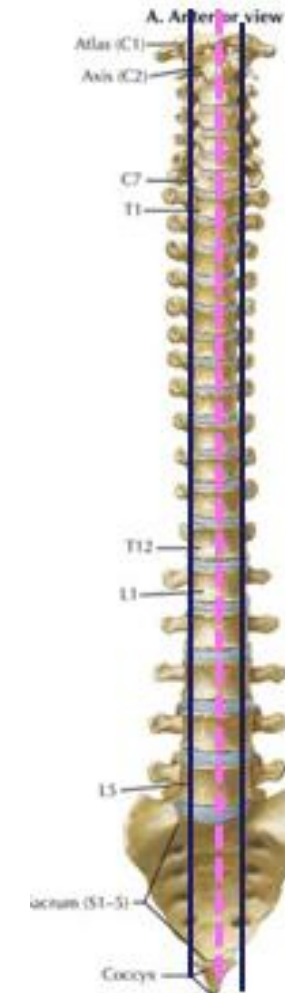
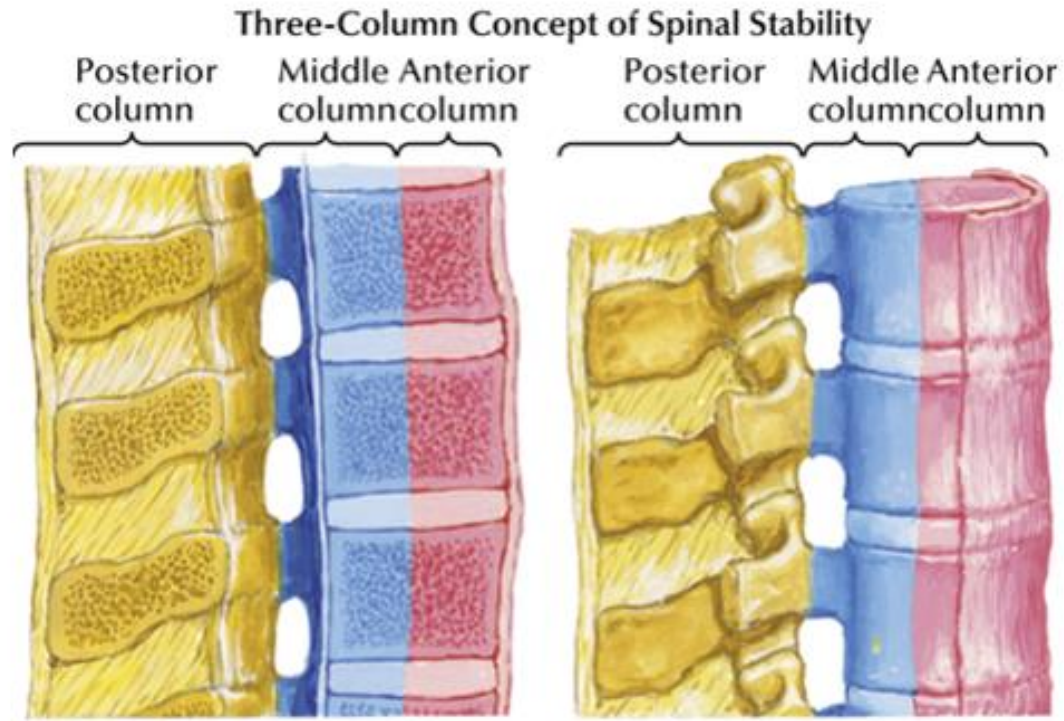


How to read an X-Ray of the Back Region

- Patient and image data
- **Patient positioning** and image quality
- Cortical outline
 - Check the cortical outline of ALL bones visible on every X-ray available
 - Check for any break in this outline
 - Do not stop when you find one abnormality - keep going until your eyes have covered all bones
- Joint spacing
 - Loss of joint space can indicate cartilage loss
- Bone and joint alignment
 - Ensure proper alignment of bone and joints relative to each other
- Bone texture
 - Look for any abnormal radio-lucencies or radio-opacities
- Soft tissues
- 2 views are better than 1
- Compare
 - With other side
 - With previous image
- **High index of suspicion - Always keep patient's clinical presentation mind**

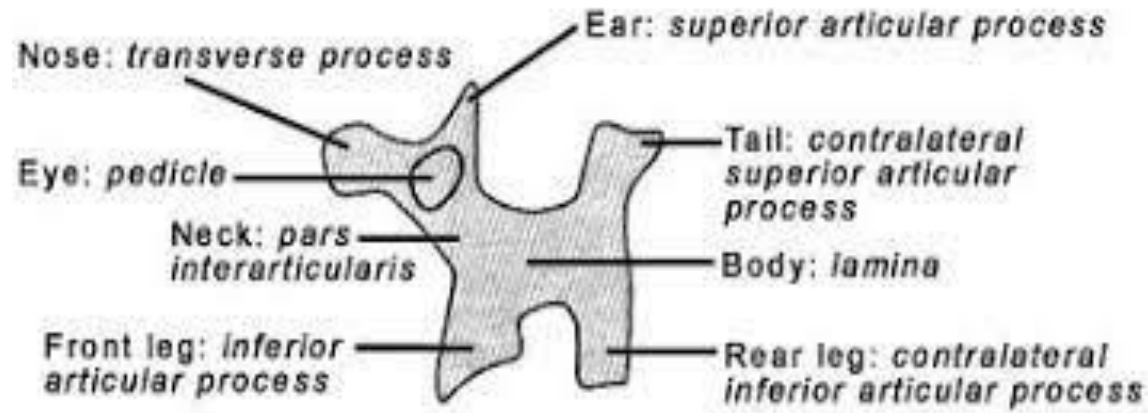
How to read an X-Ray of the Back Regain

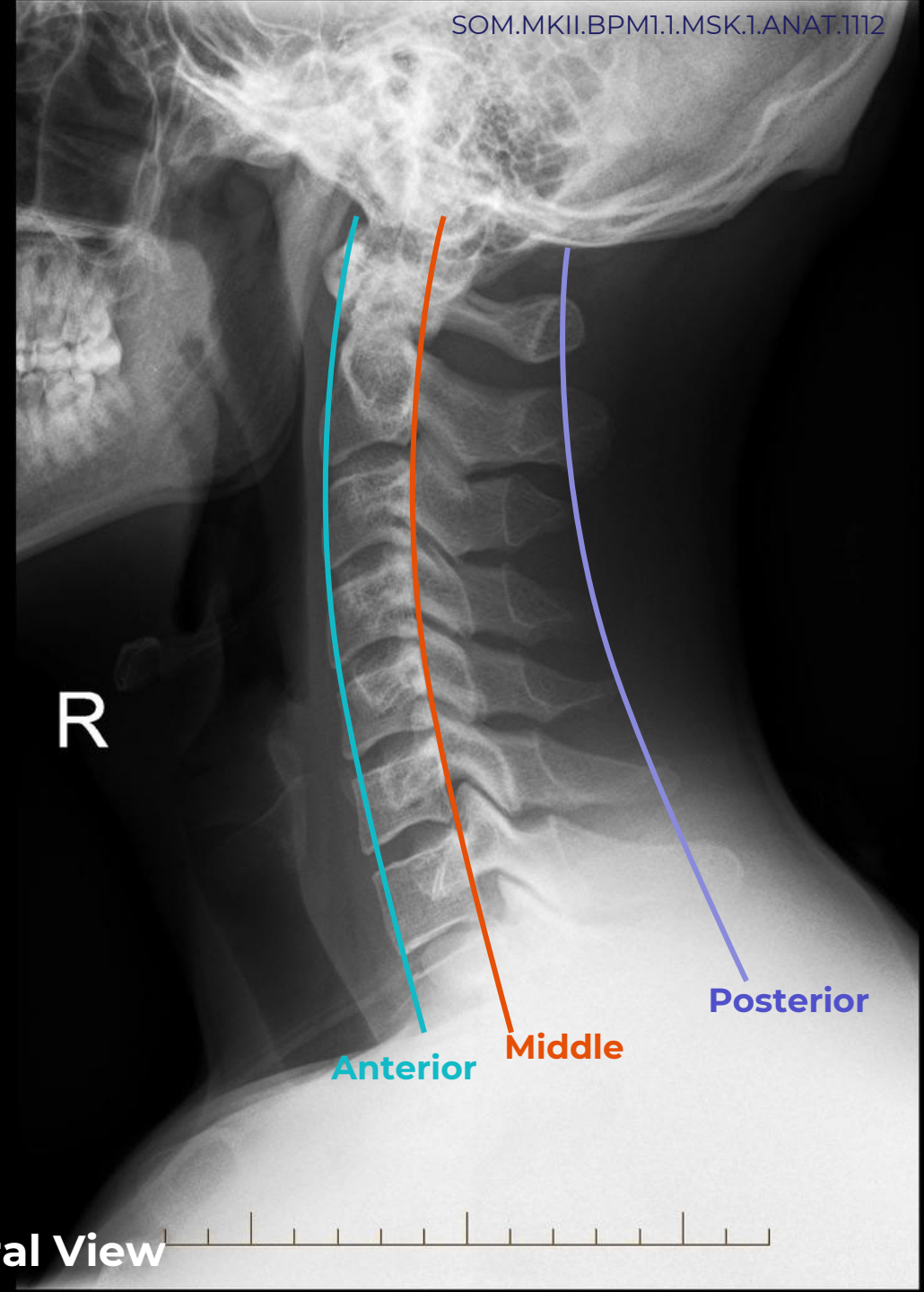
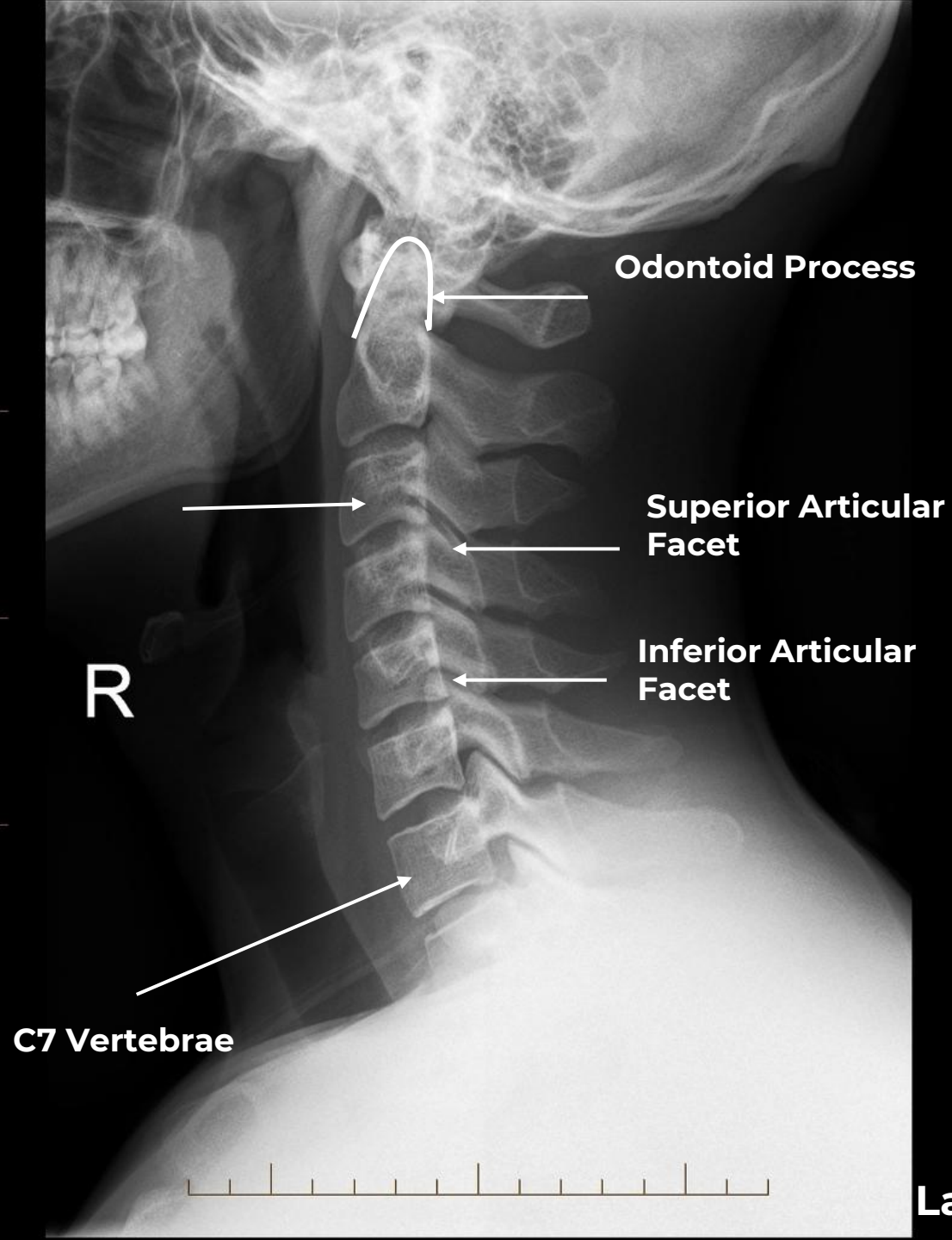
- Lateral view “Three Column” approach
 - Anterior Longitudinal Line
 - Posterior Longitudinal Line
 - Spinolaminar Line
- Anterior-Posterior view
 - Lateral Lines (2)
 - Spinous Line



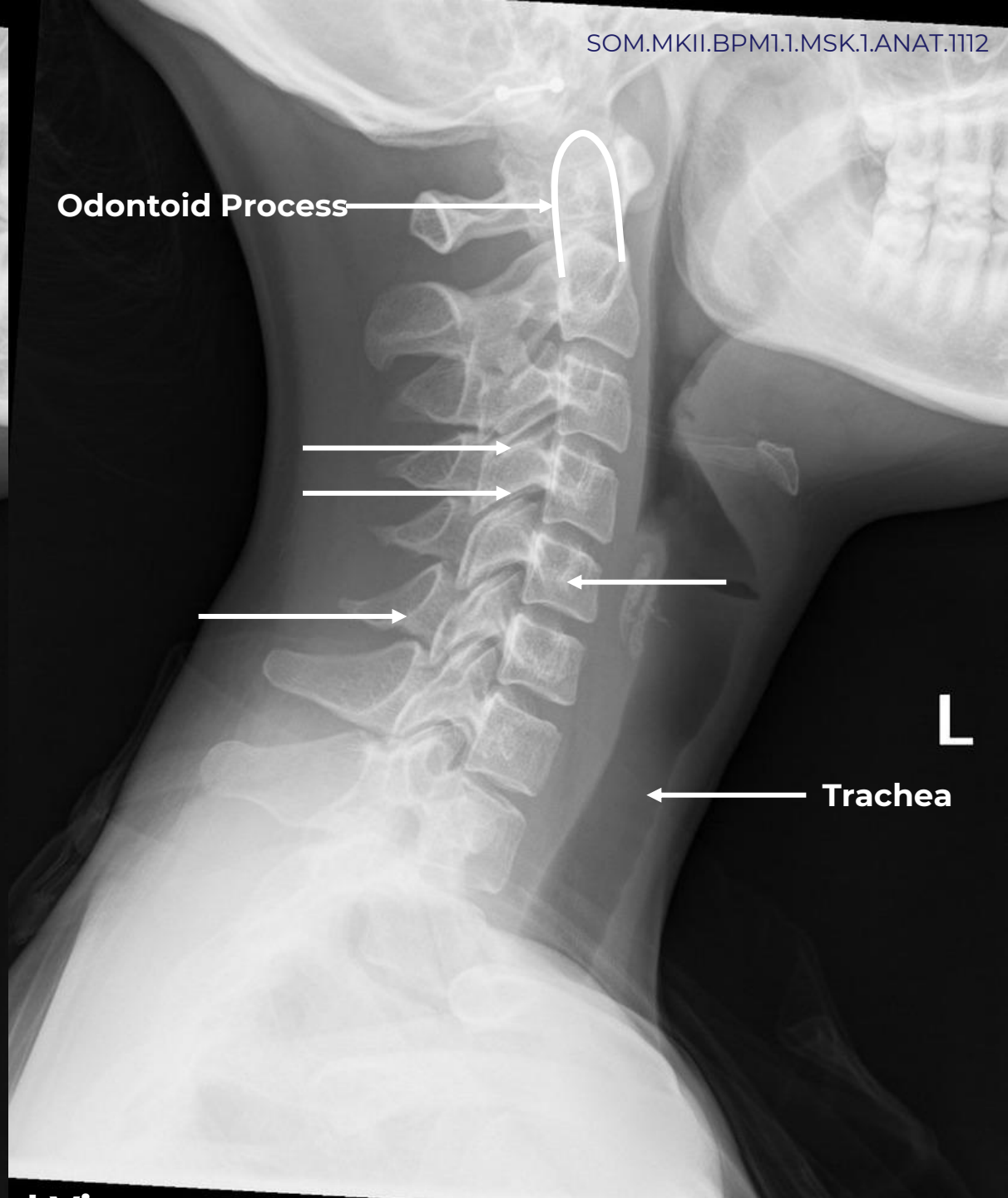
How to read an X-Ray of the Back Region

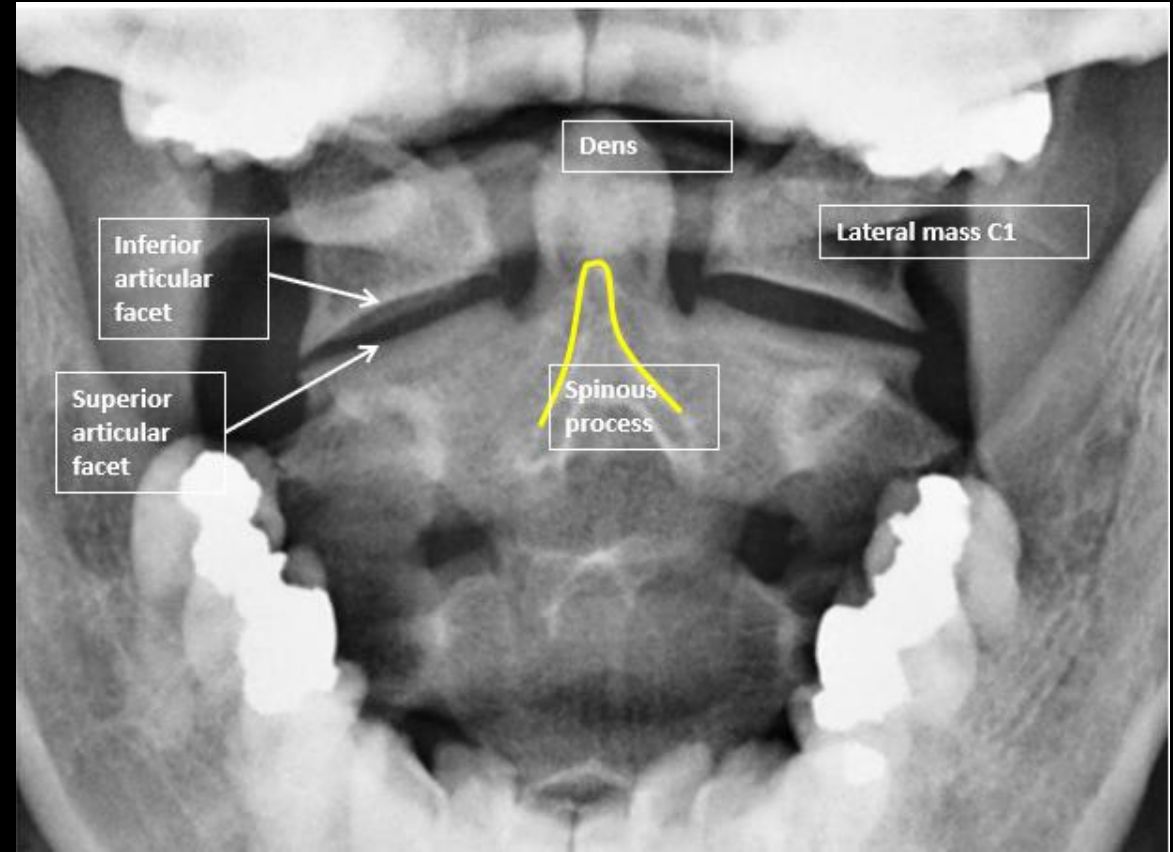
- Oblique View
 - **Scotty Dog – Lumbar region**
- Open Mouth View
 - Alignment of odontoid process
 - with C1 Lateral Masses

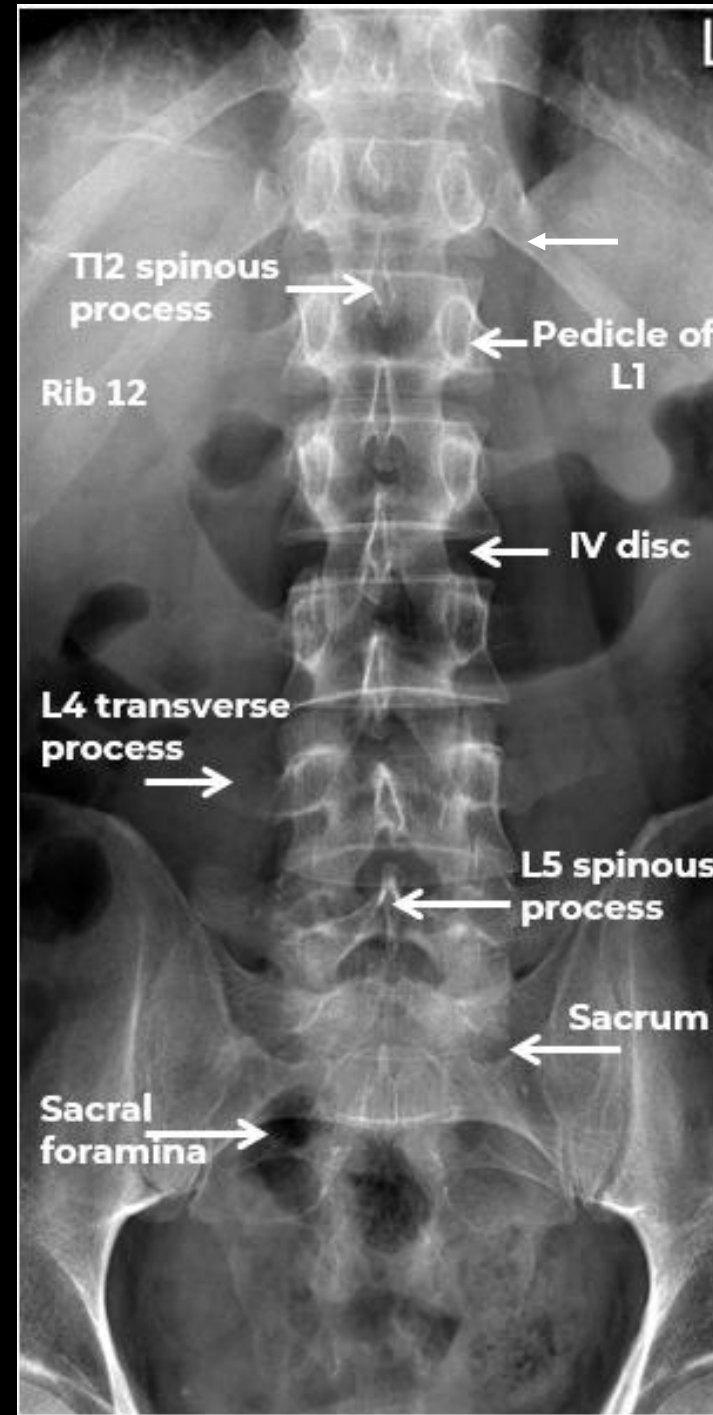




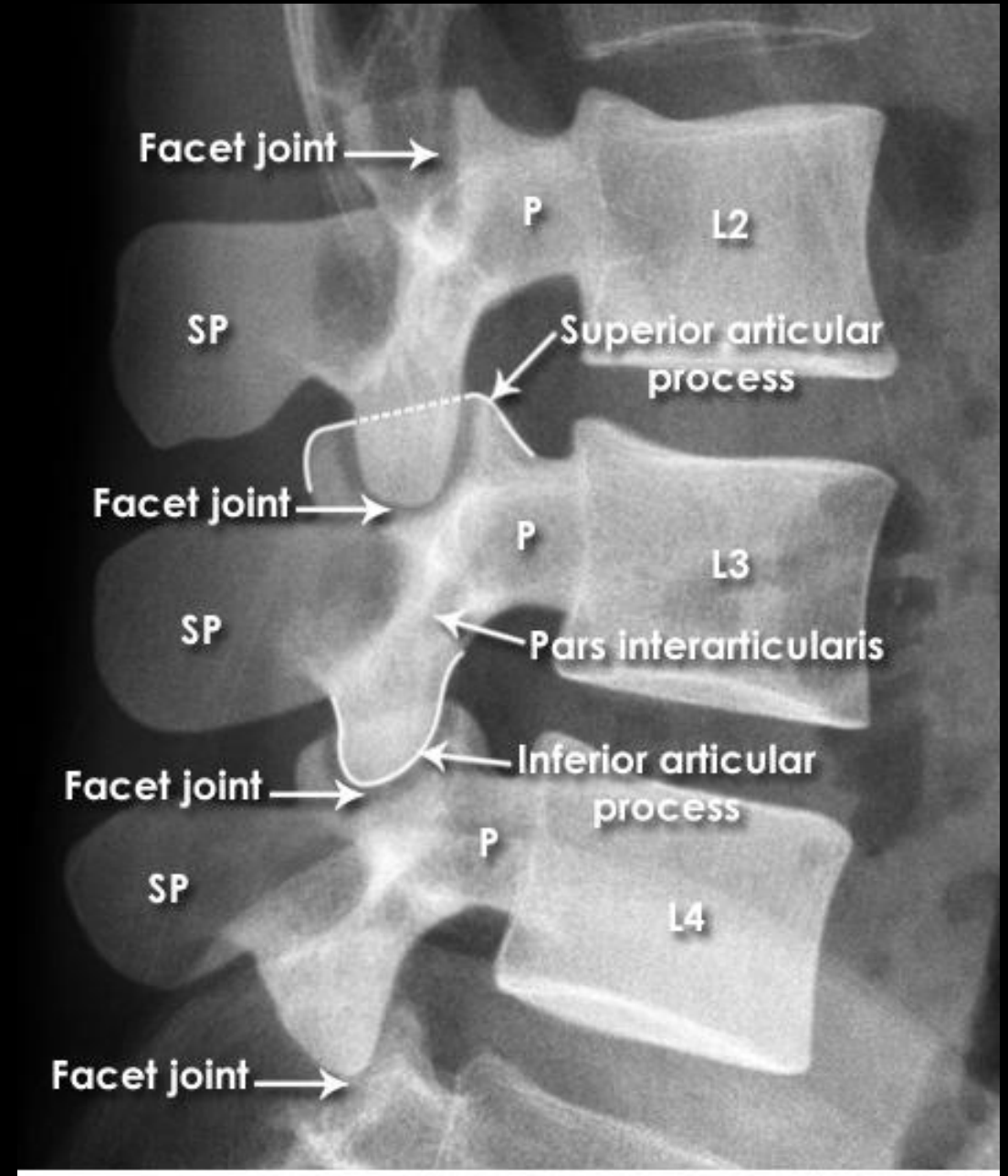
Lateral View



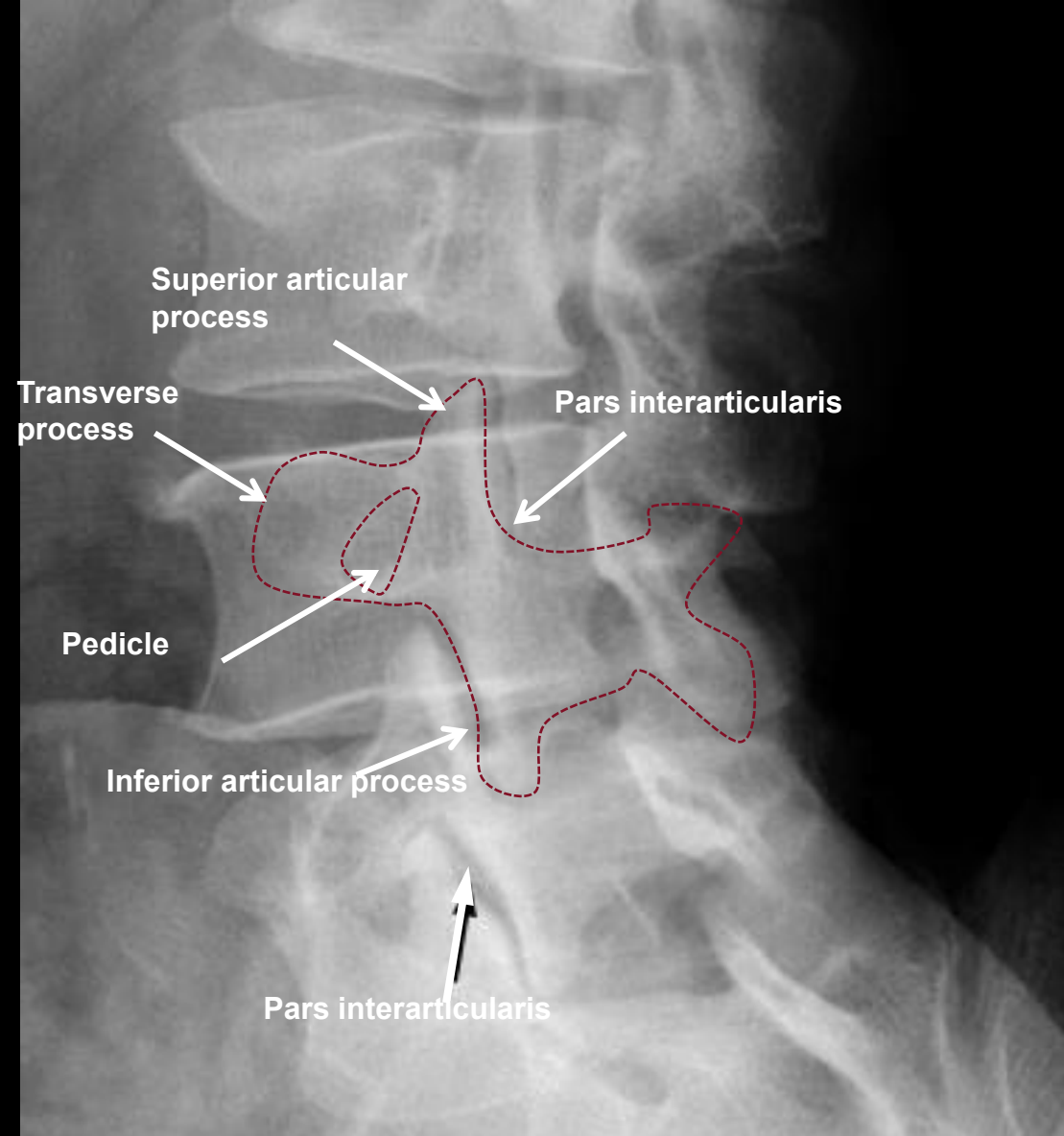
**Odontoid View**



Frontal View



Lateral View



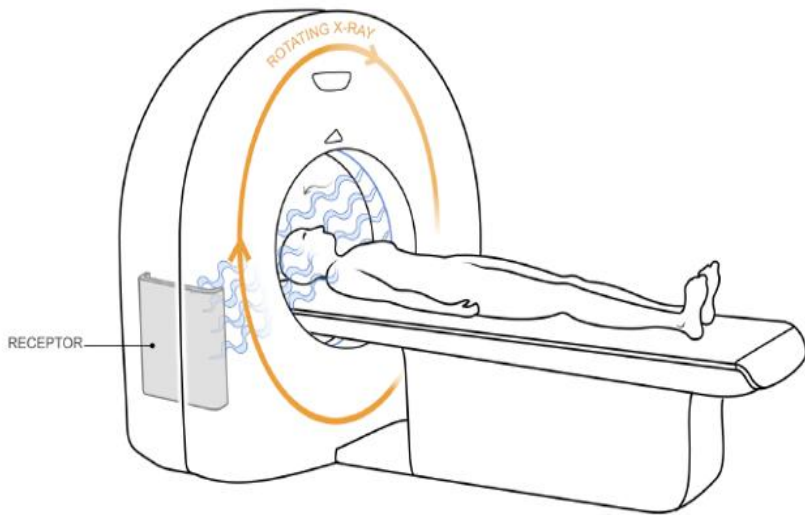
Lateral- Oblique View

CT Scan Computed Tomography

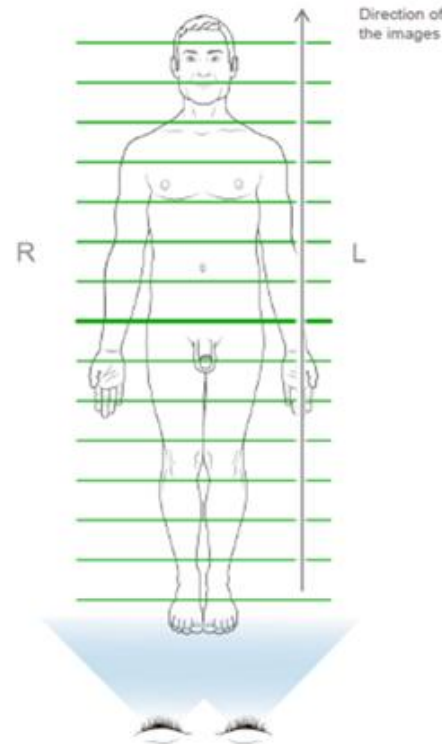
- Also utilizes X Ray Images. X-Ray Beams are rotated about the patient
- Images are taken from different angles, computer processing then creates cross-sectional images (slices)
- **2D Image** is created
- Creation of an image in different planes (Axial, Coronal, or Sagittal)

Usually followed
by an X-Ray in the
case of Trauma.

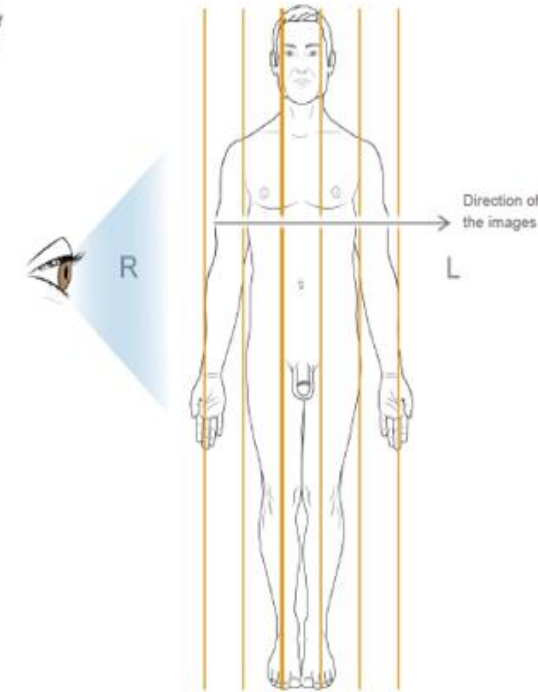
Better
visualization of
soft tissue



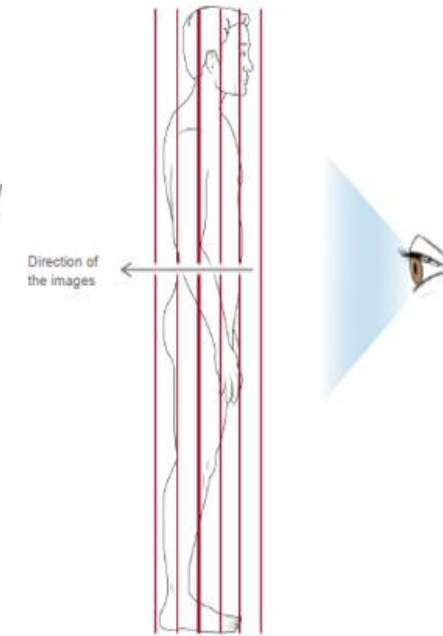
AXIAL PLANE



SAGITTAL PLANE

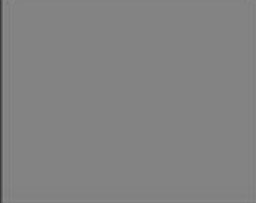
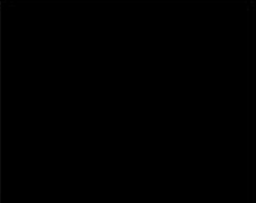


CORONAL PLANE



CT Scan Computed Tomography

- Images are still in shades of gray
- **BONE and Metal IS BRIGHT WHITE!**
- Better definition of soft tissue
- Enhances the appearance of traumatic injuries that were not visible on X-Ray
- Images can be enhanced with window views
 - Manipulation of grey scale to enhance structures
- Now image reconstruction can create 3D Images
- Can utilize **contrasts**
 - CT Scan without Contrast vs. CT Scan with Contrast

 COMPACT BONE	 Bright white
 MUSCLE	
 FLUID	
 FAT	
 BONE MARROW	



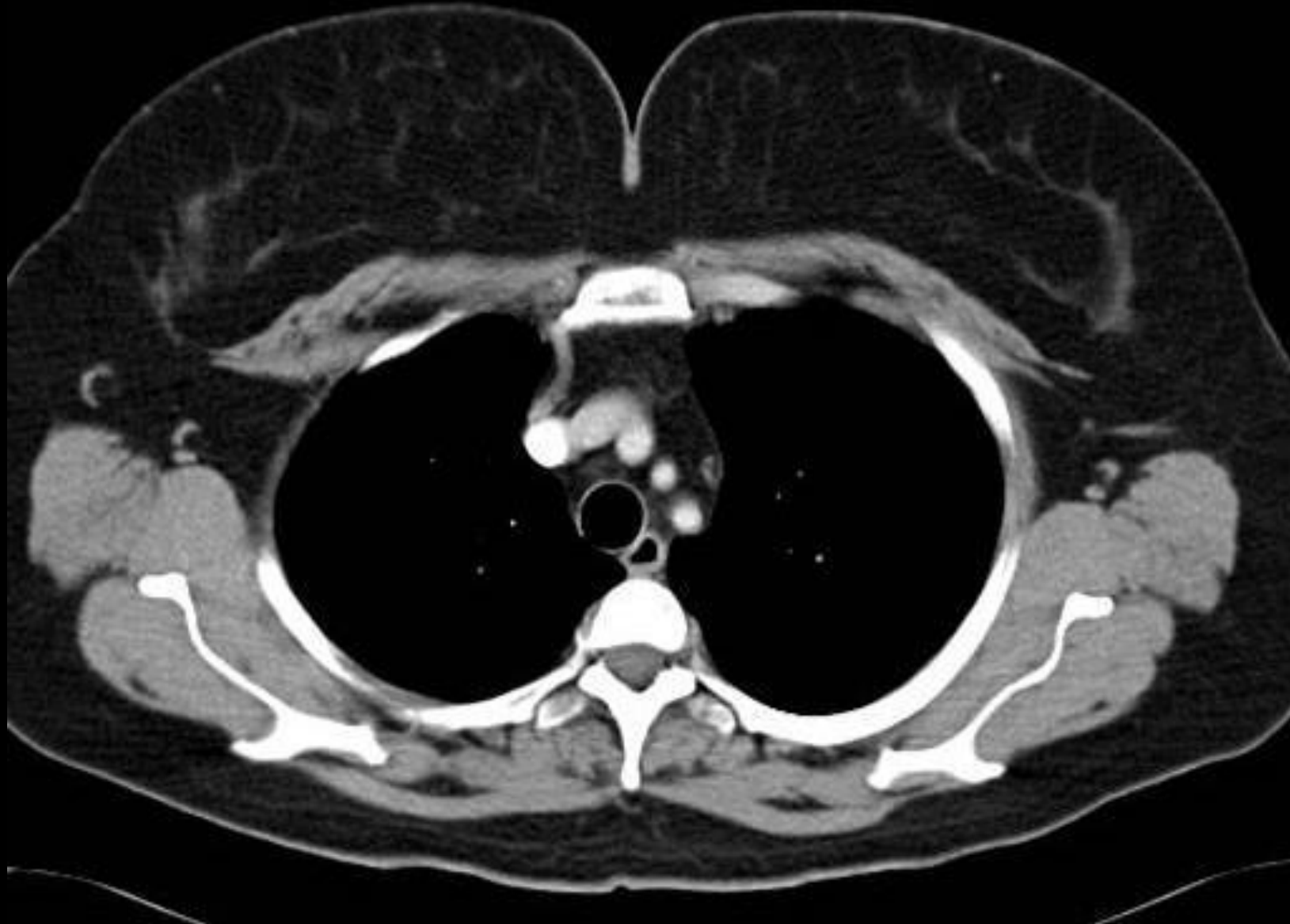
Intervertebral
Disc Space

Vertebral
Body

Sacral
Promontory

Spinous
Process

Lateral (Midsagittal)



Transverse Axial View

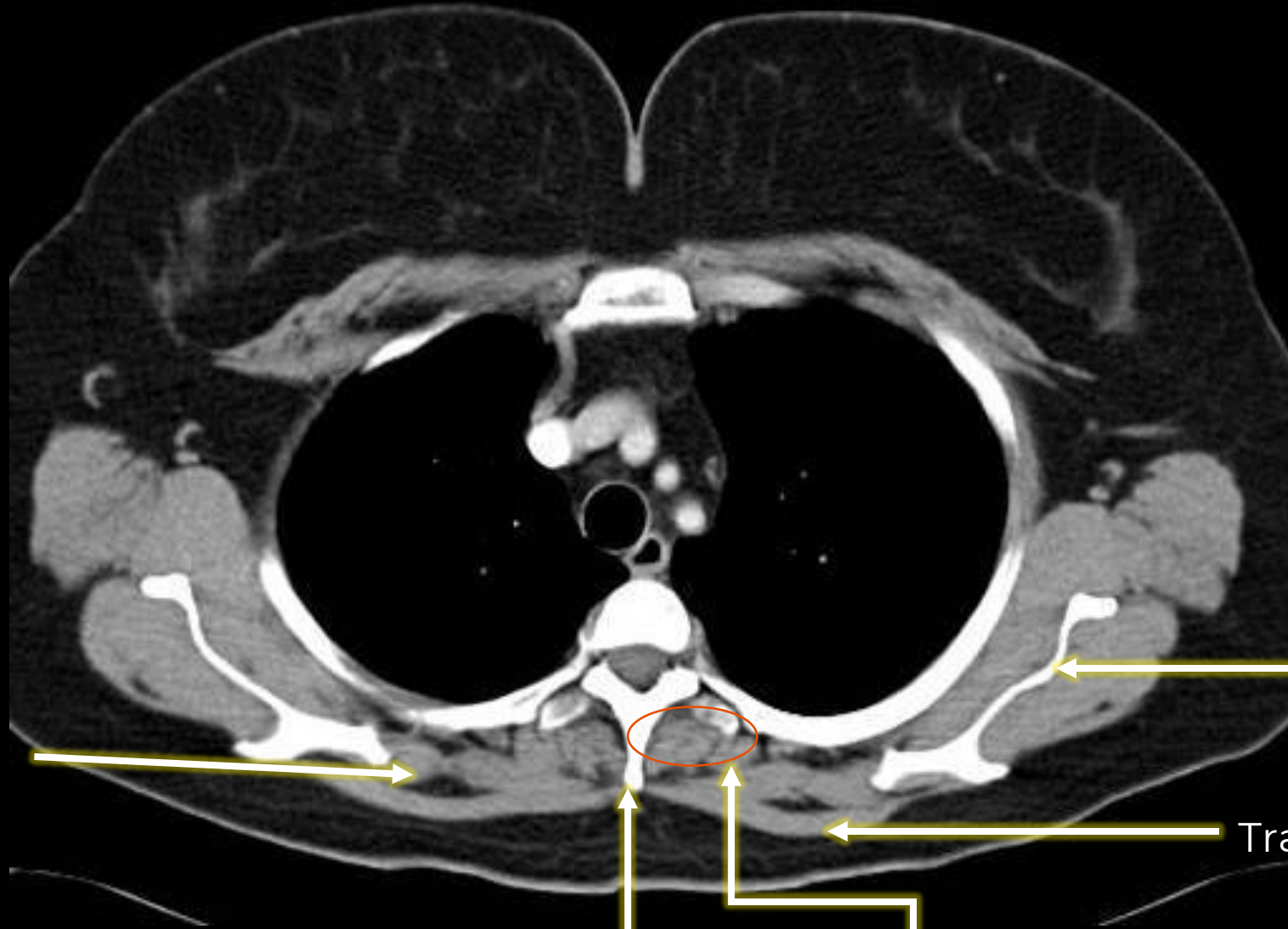
Rhomboids

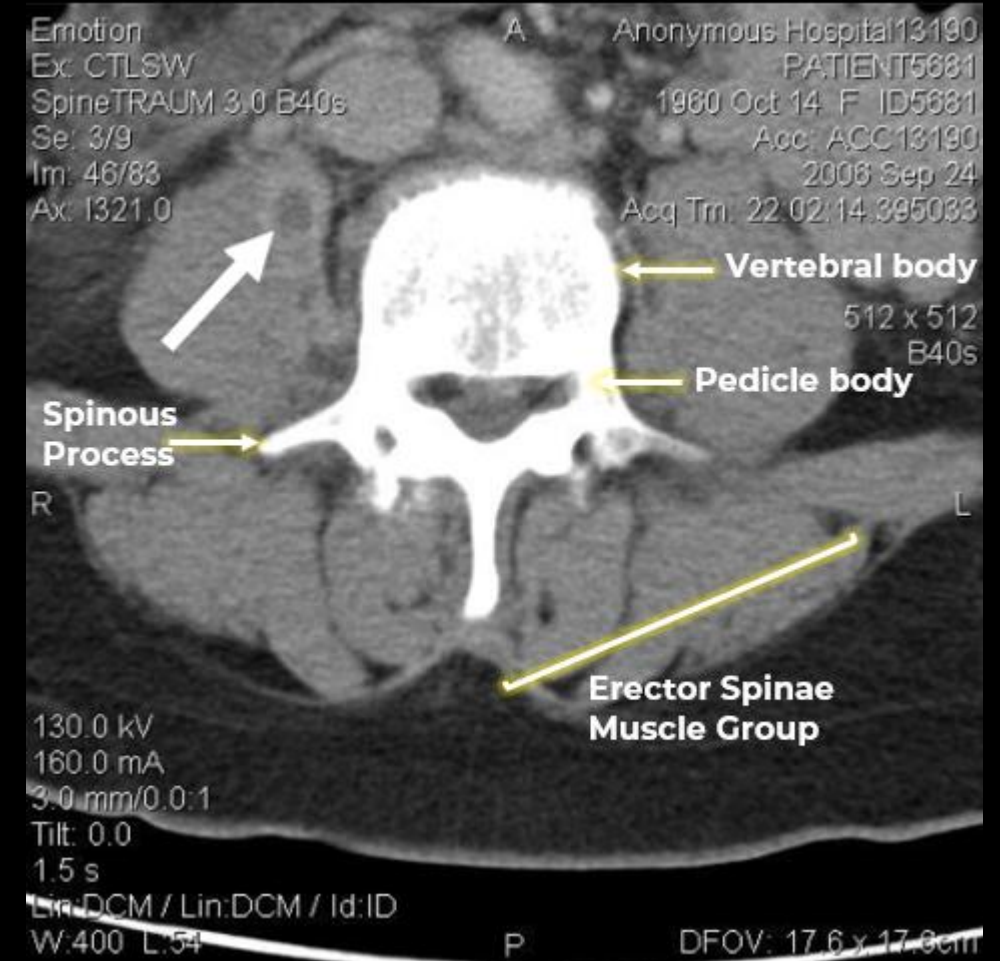
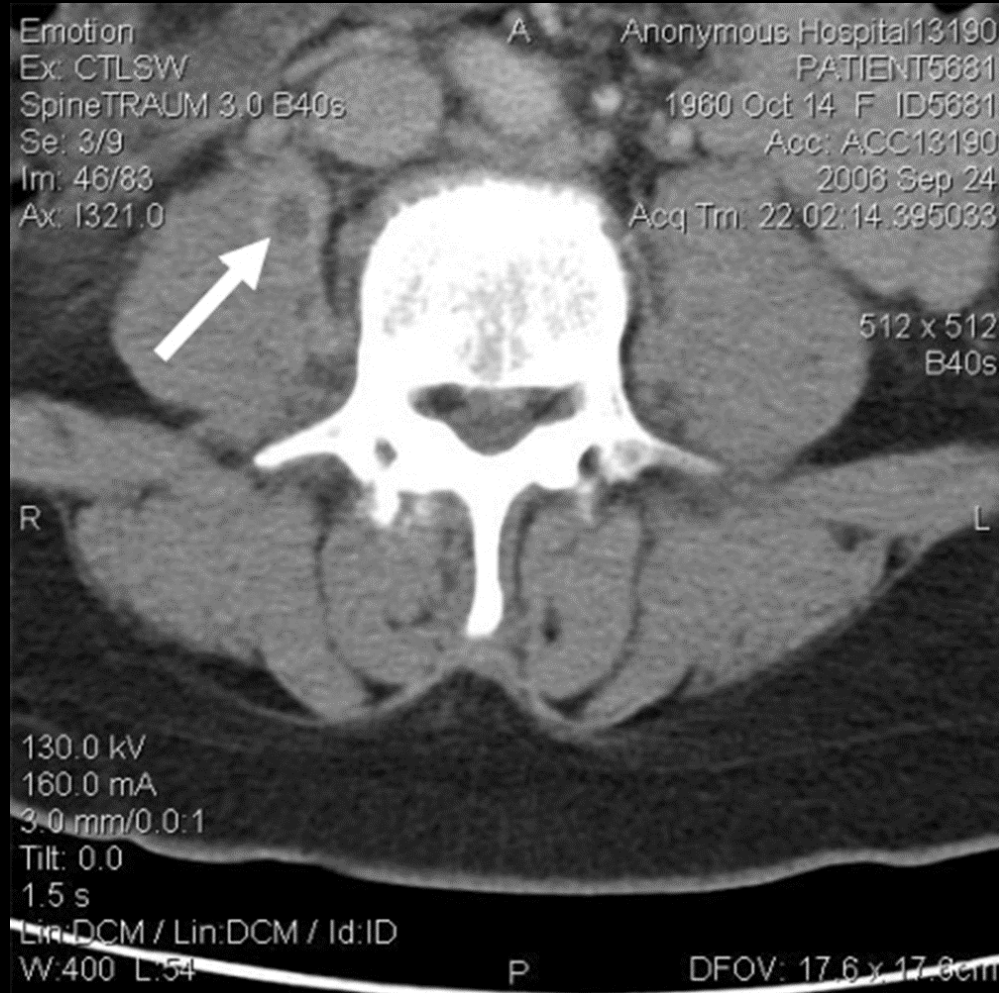
Scapula

Trapezius Muscle

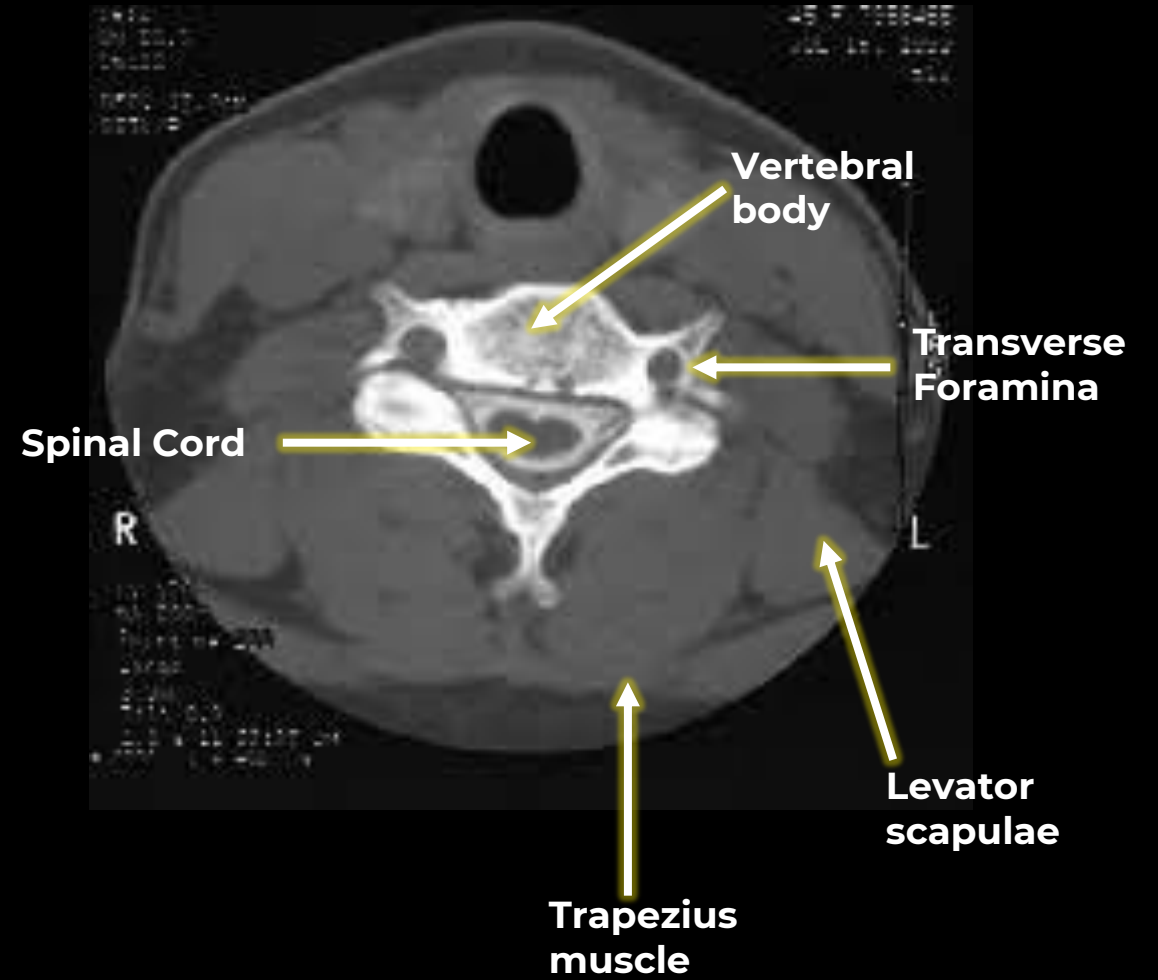
Spinous Process

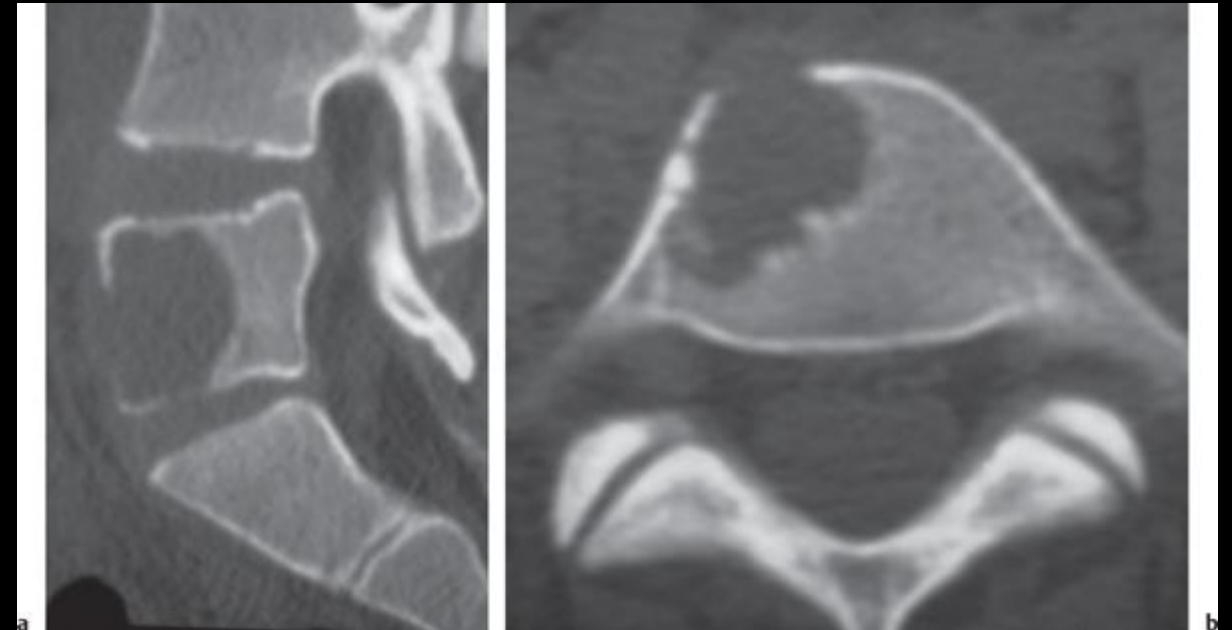
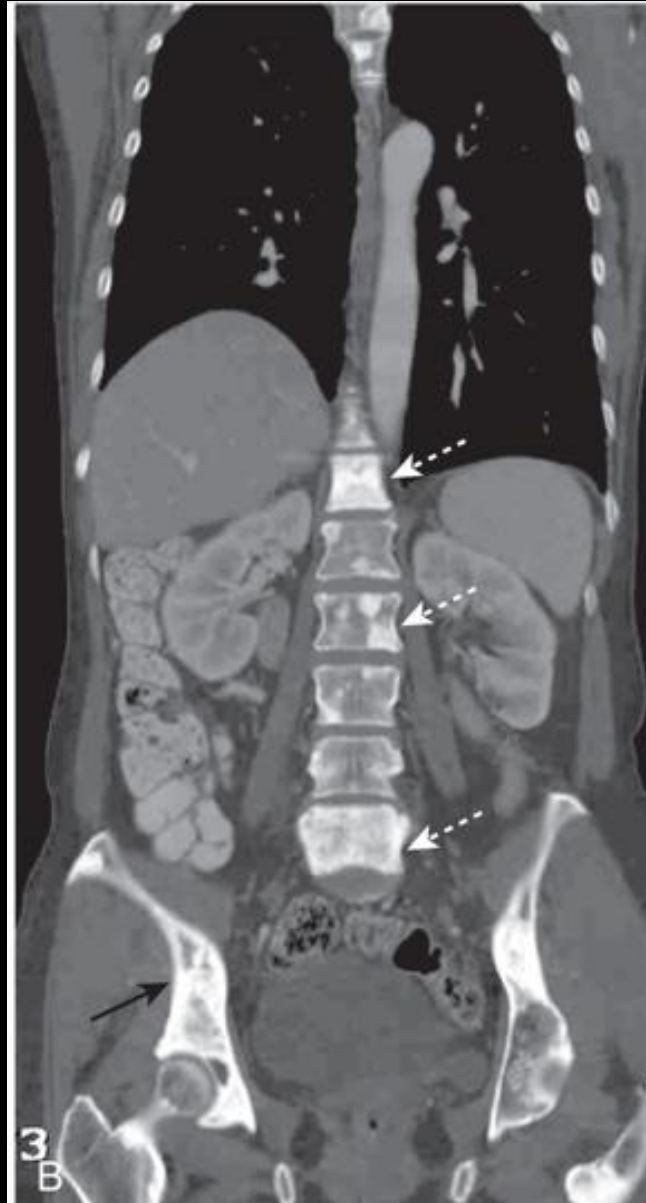
Erector Spinae Muscles





Transverse Axial View

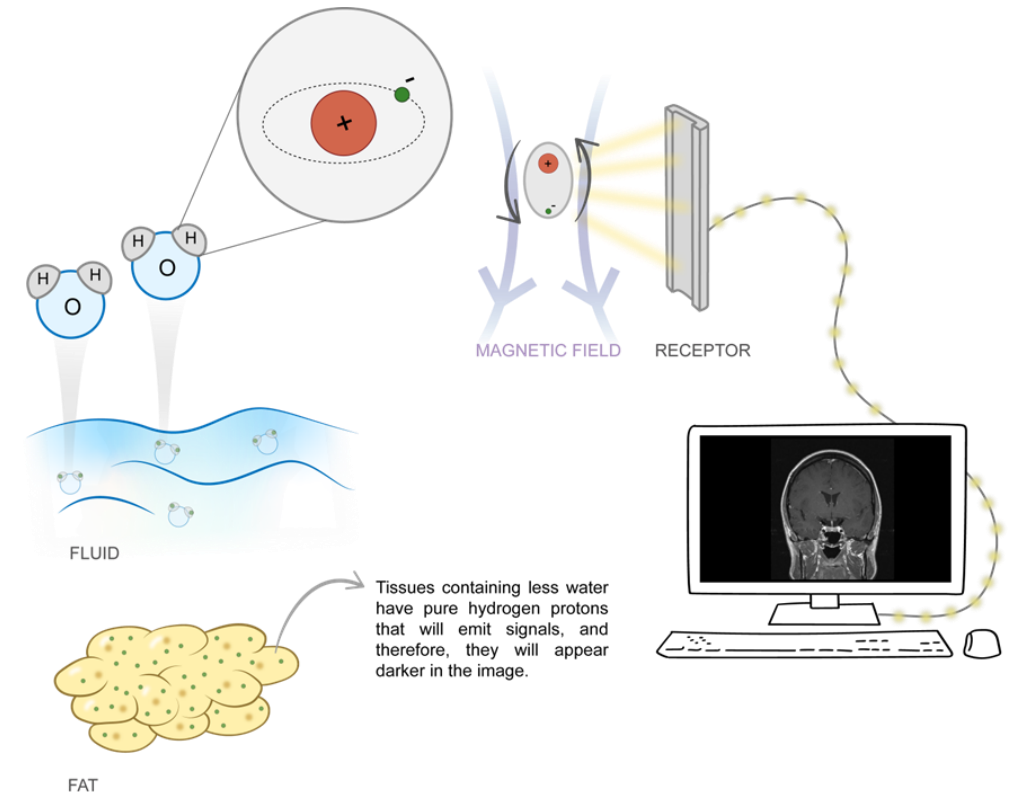
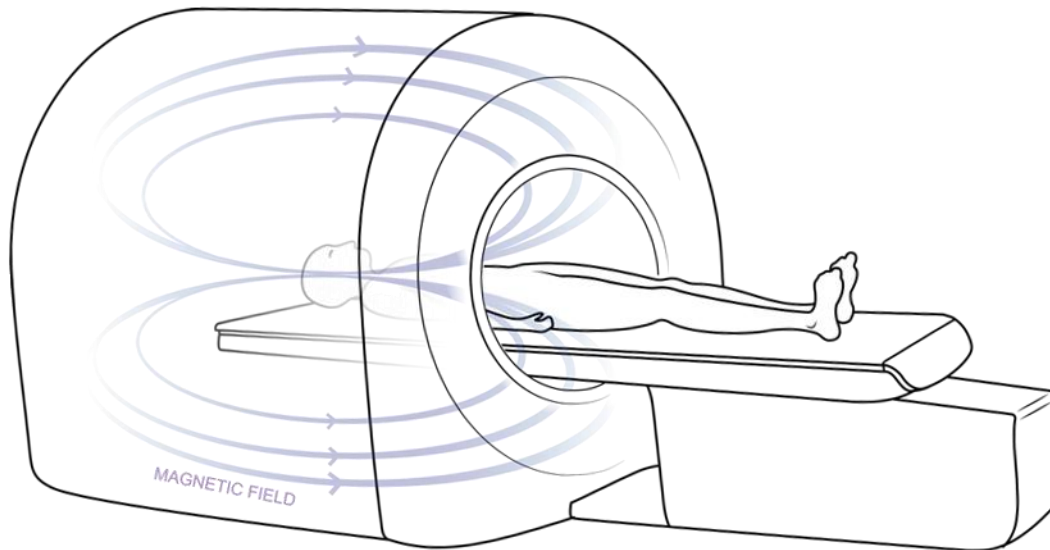
**Transverse Axial View**



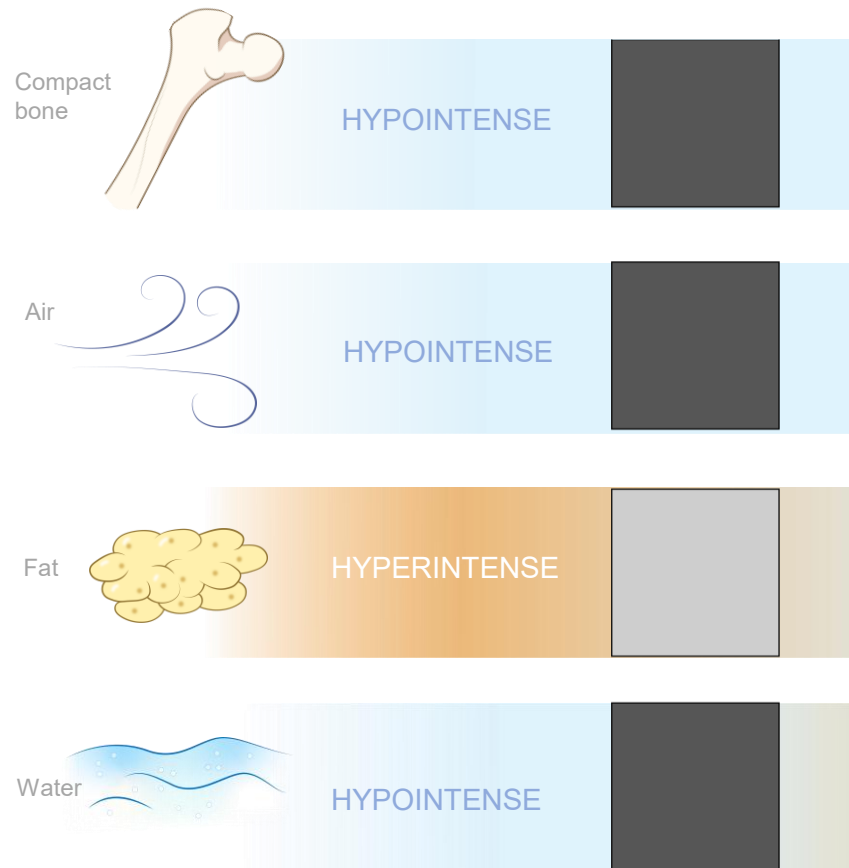
What view is this?

MRI

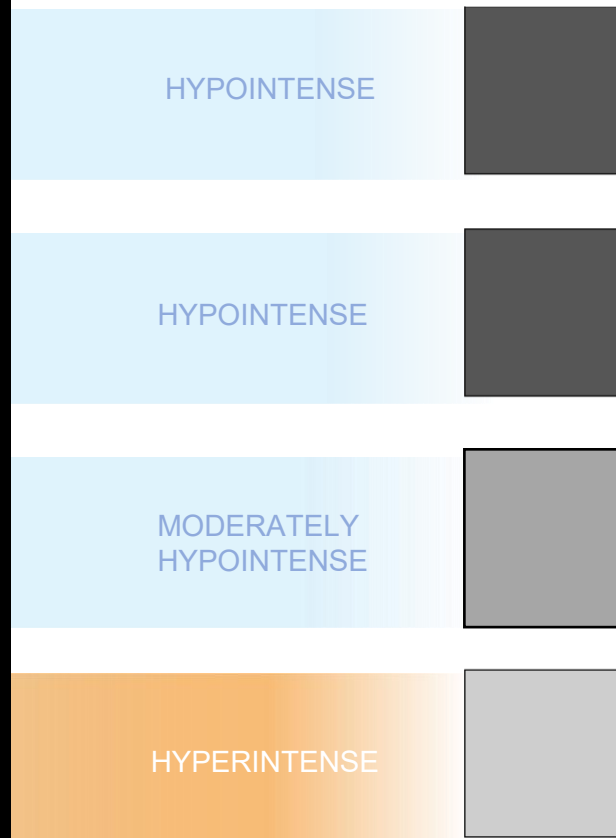
- MRI uses the magnetic properties of spinning hydrogen atoms
- MRI signal depends on many varied properties of the tissues and structures:
 - Number of hydrogen atoms present in tissue (proton density)
 - Chemical environment of the hydrogen atoms, e.g. whether in free water or bound by fat
 - Flow: blood vessels or CSF
 - Magnetic susceptibility: T1 relaxation time and T2 relaxation time
- Patient is placed into a magnetic field



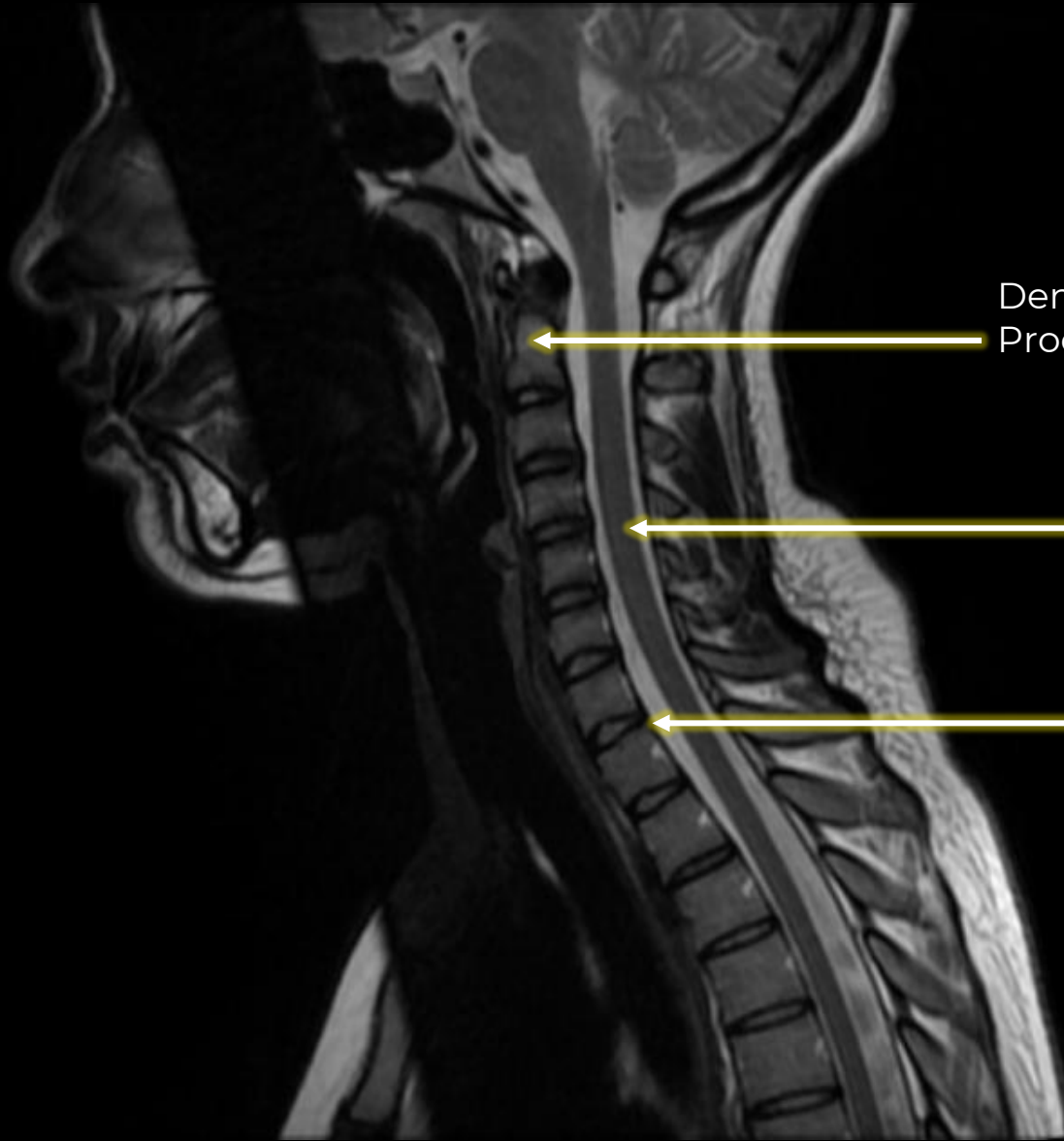
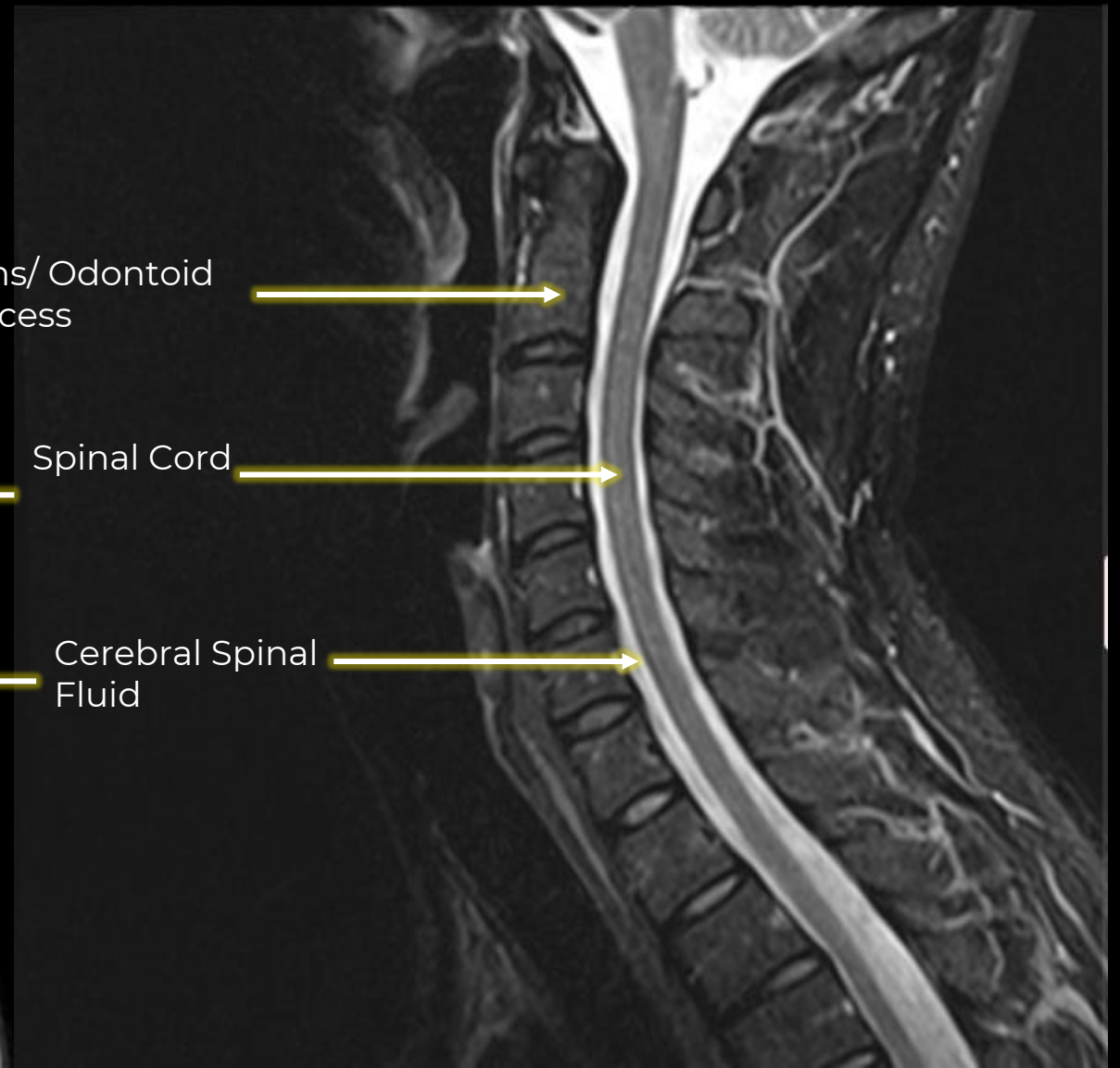
T1 vs T2 Weighted MRI Images

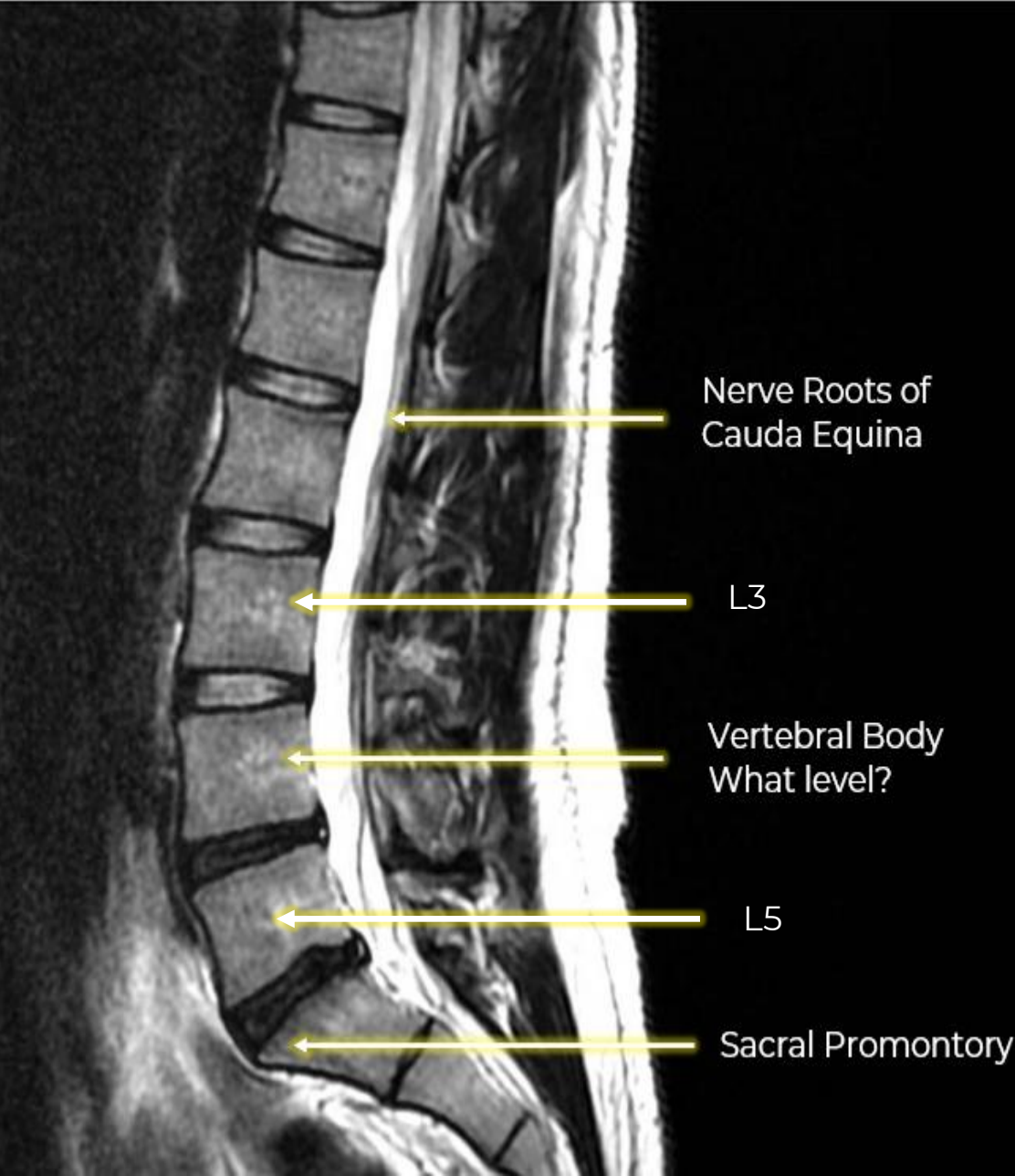


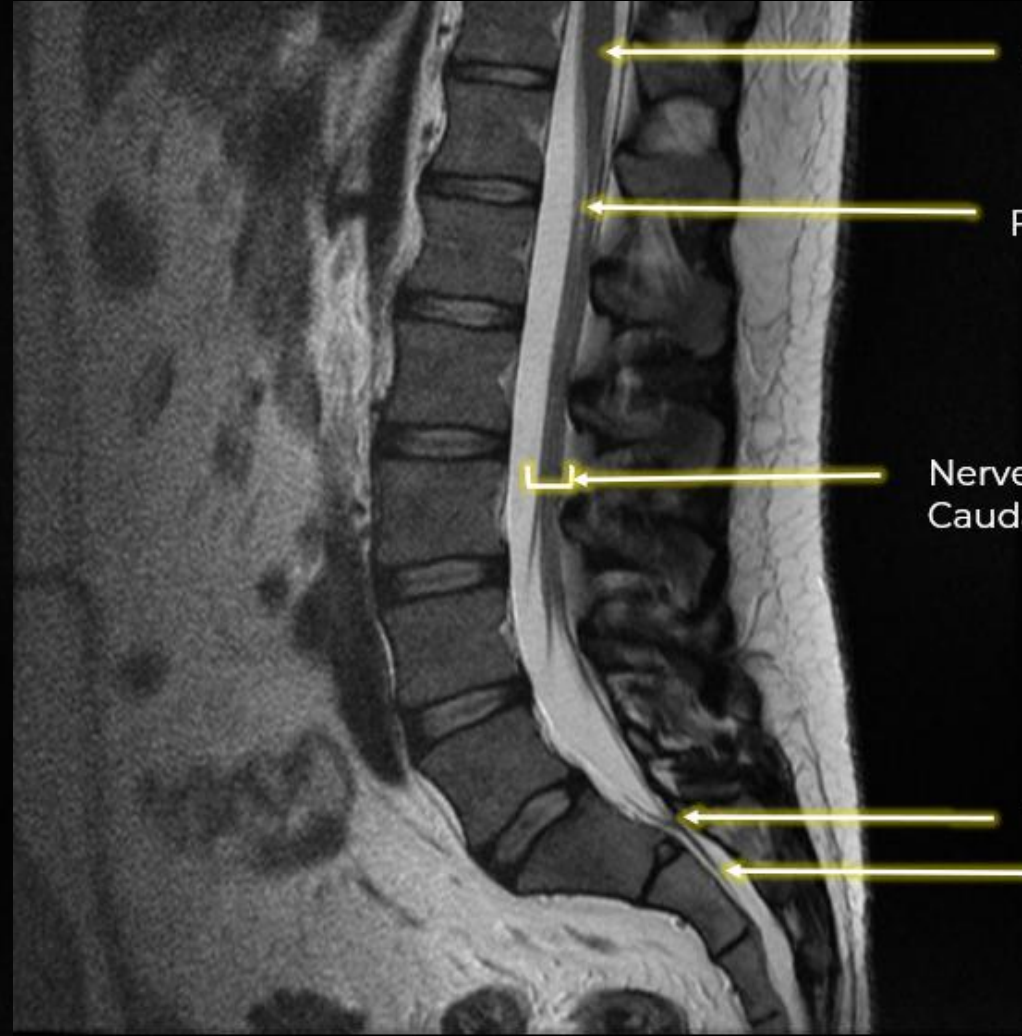
Sagittal MRI of the spine. Note the dark cerebrospinal fluid (CSF)



Sagittal MRI of the spine. Note the bright cerebrospinal fluid (CSF)

**T1 Weighted MRI****T2 Weighted MRI**





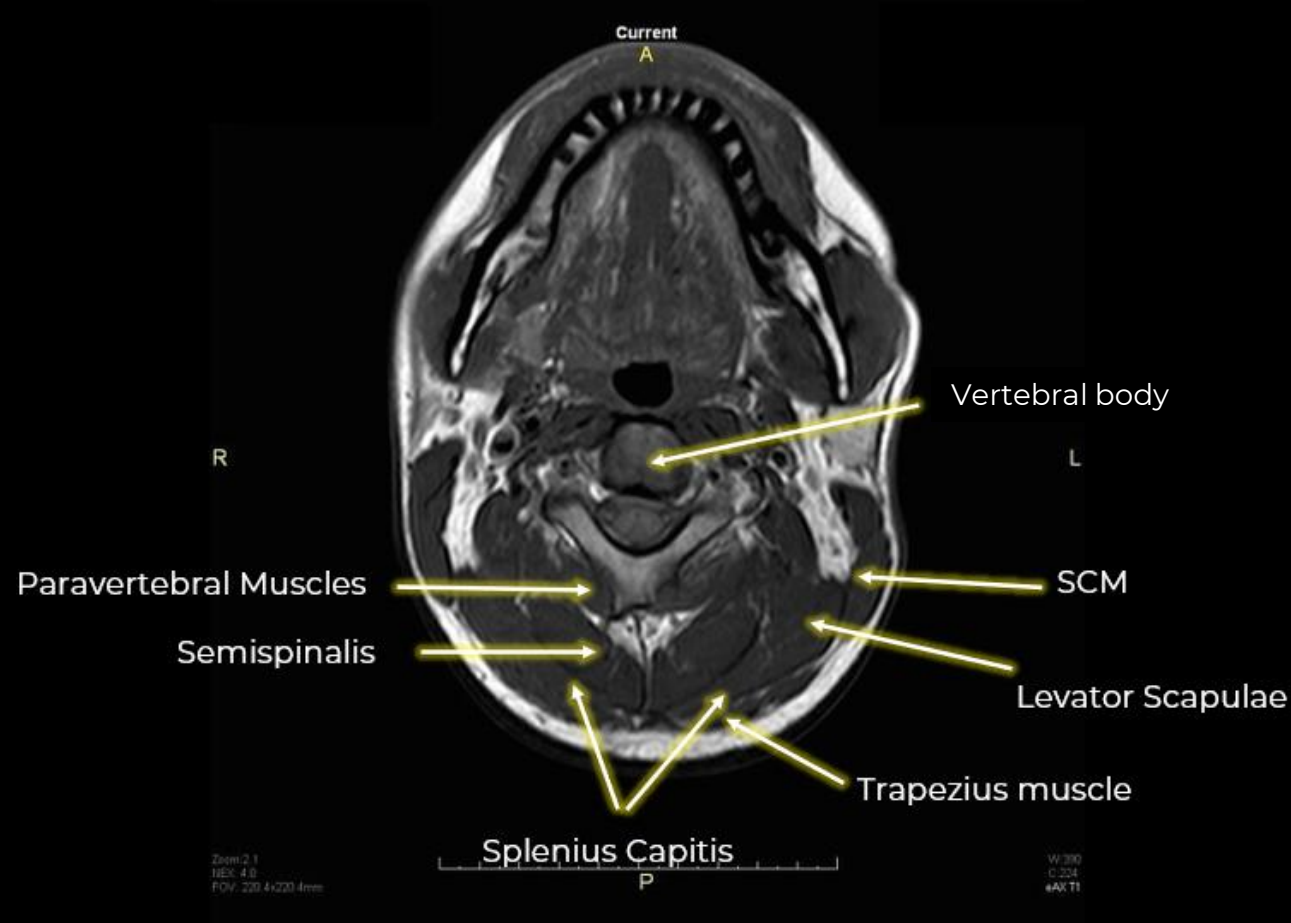
Conus medullaris

Filum Terminale

Nerve Roots of
Cauda Equina

Termination of Dura Sac

Filum Terminale

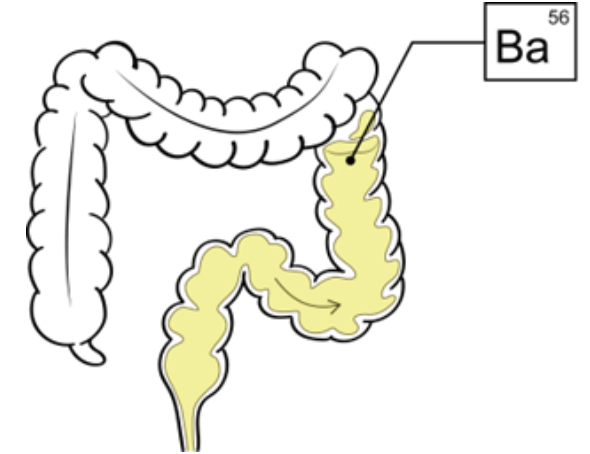




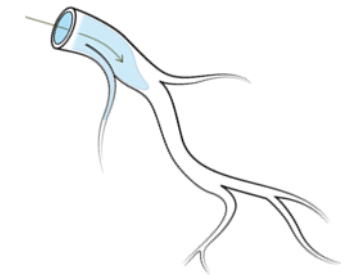
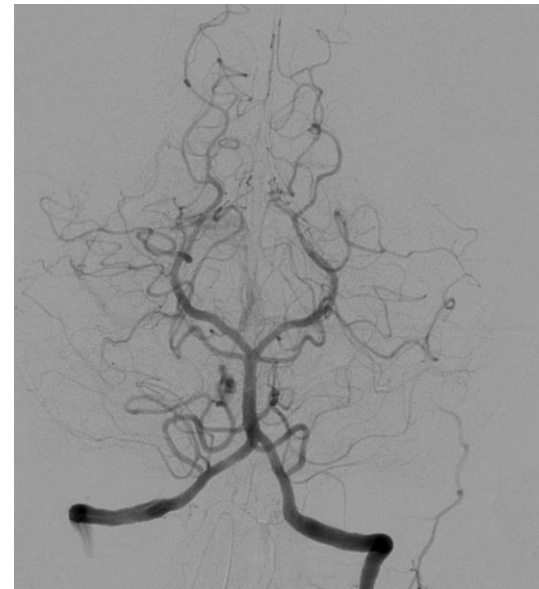
Erector Spinae
Muscle Group

Contrast Studies

- Necessary to analyze structures that do not have inherent contrast differences from the surrounding tissues e.g. abdominal structures, blood vessels
 - in general, a contrast agent is a pharmaceutical substance which is radiopaque (attenuates x-rays and therefore shows up as white)
-
- CT Scan with Contrast
 - Fluoroscopy
 - CT Angiograms
 - Barium Enema
 - Barium Swallow



Barium Enema (Gastrointestinal Tract)



Subtraction angiography (Circulatory System)

Comparing Images – Complete When Studying

	X-RAY	CT-Scan	MRI (T1)	MRI (T2)
AIR				
FAT				
FLUID				
BONE				

A WINNER
IS A
DREAMER
WHO
NEVER
GIVES UP

Questions email:
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Or
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